



ASSESSING MENTAL HEALTH USING FACIAL EMOTION DETECTION

PROBLEM DEFINITION

STRESS

Society, Family, Work



Mental Health

Depression, Anxiety



Emotions

Facial Expression, Voice



Depression results in global
economic loss of ~1 Trillion
US\$/year

PROBLEM TO SOLVE

- How can we detect and treat depression?
- Can we use facial expression to detect emotions and predict depression/anxiety?
- What are some policies that can be adopted to treat depression/anxiety?



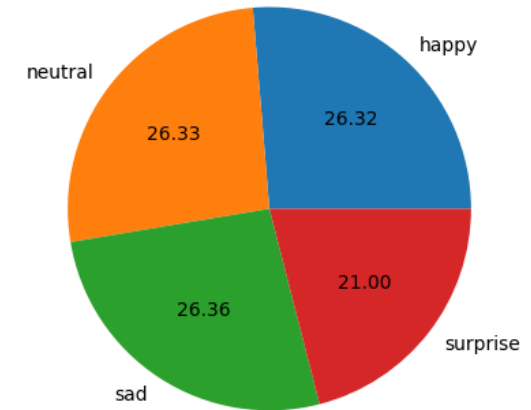
SOLUTION APPROACH

- Data Exploration led to key takeaways.
- FER 2013 Data Set consists of 48x48 pixel grayscale images of faces.
- The face is mostly centered and occupies about the same amount of space in each image.
- Some images have
 - facial tilt, rotation
 - hands over mouth/ears
 - Eyes behind glasses/shades



SOLUTION APPROACH (contd.)

- 4 Emotions (Class)
 - Happy
 - Neutral
 - Sad
 - Surprise
- All classes fairly uniformly distributed (Surprise 21% - lowest)



SMALL TRAINING SET

- Data Augmentation
 - Rotation
 - Horizontal Flip
 - Shear



PROPOSED MODEL SOLUTION

Convolutional Neural Network based model chosen to predict facial emotions

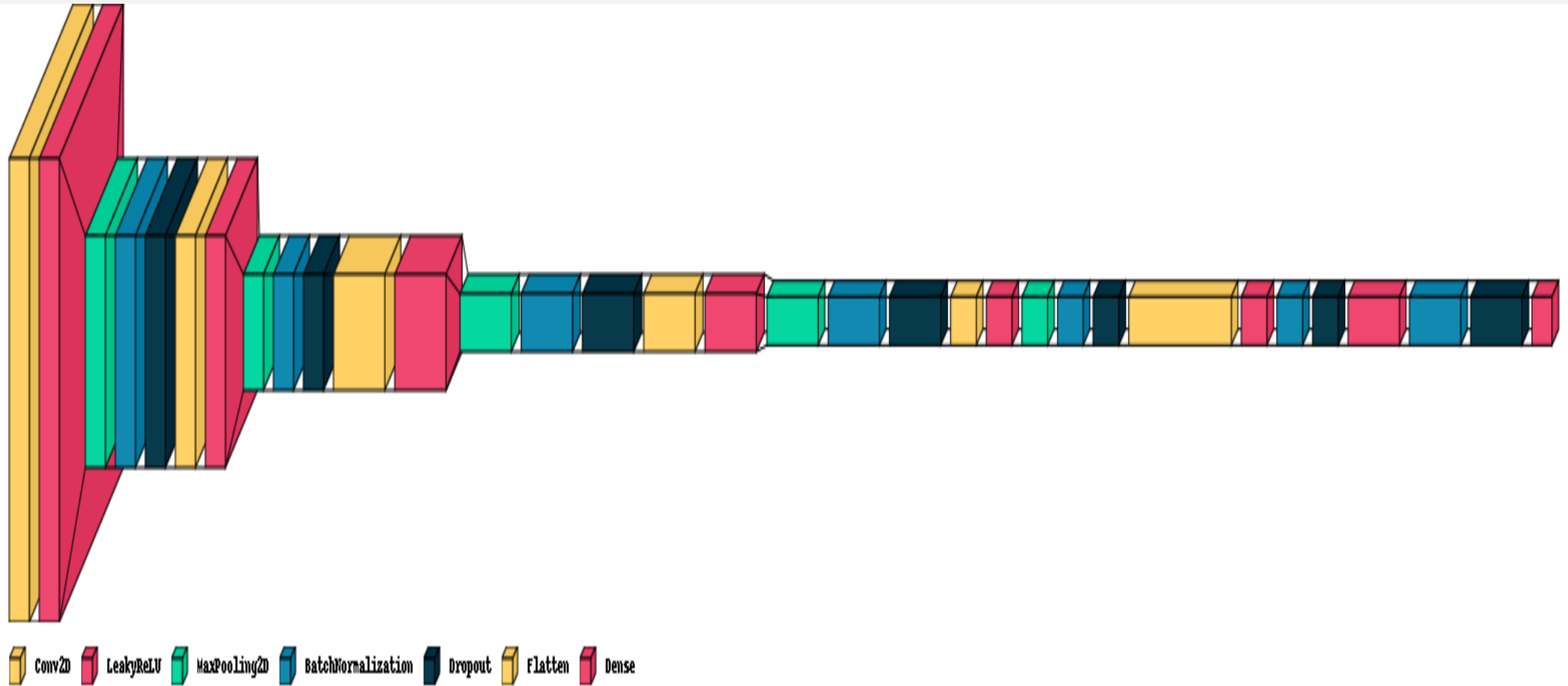
Feature extraction is done using 5 convolutional layers.

Classification is done using 2 Dense layers.

Hyper Parameters: Filter size, Learning rate, Batch size optimized for performance (Time constraints)

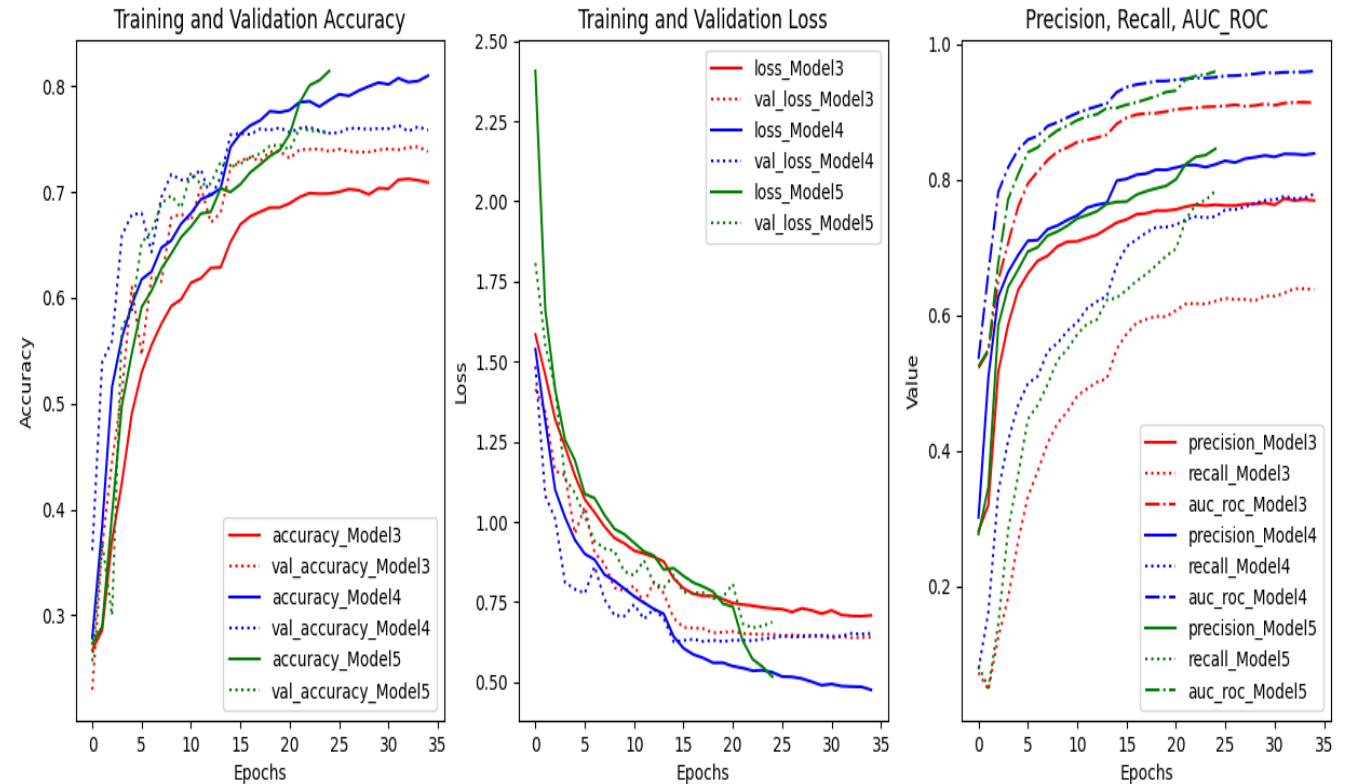
Regularization: MaxPooling, L2 kernel regularizer, Dropout, BatchNormalization

Final Model



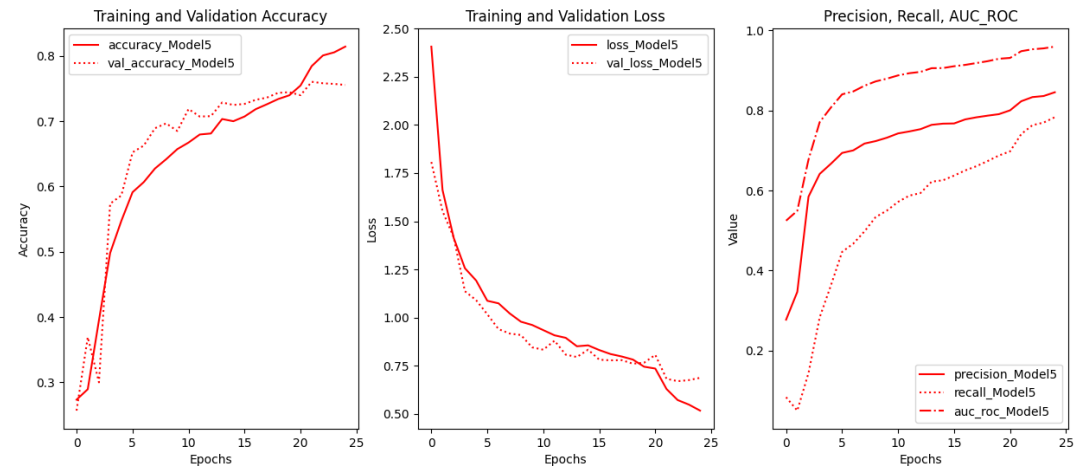
Model Design Iteration

- Model 3 – Underfitting
- Model 4 – Overfitting
- Final Model 5

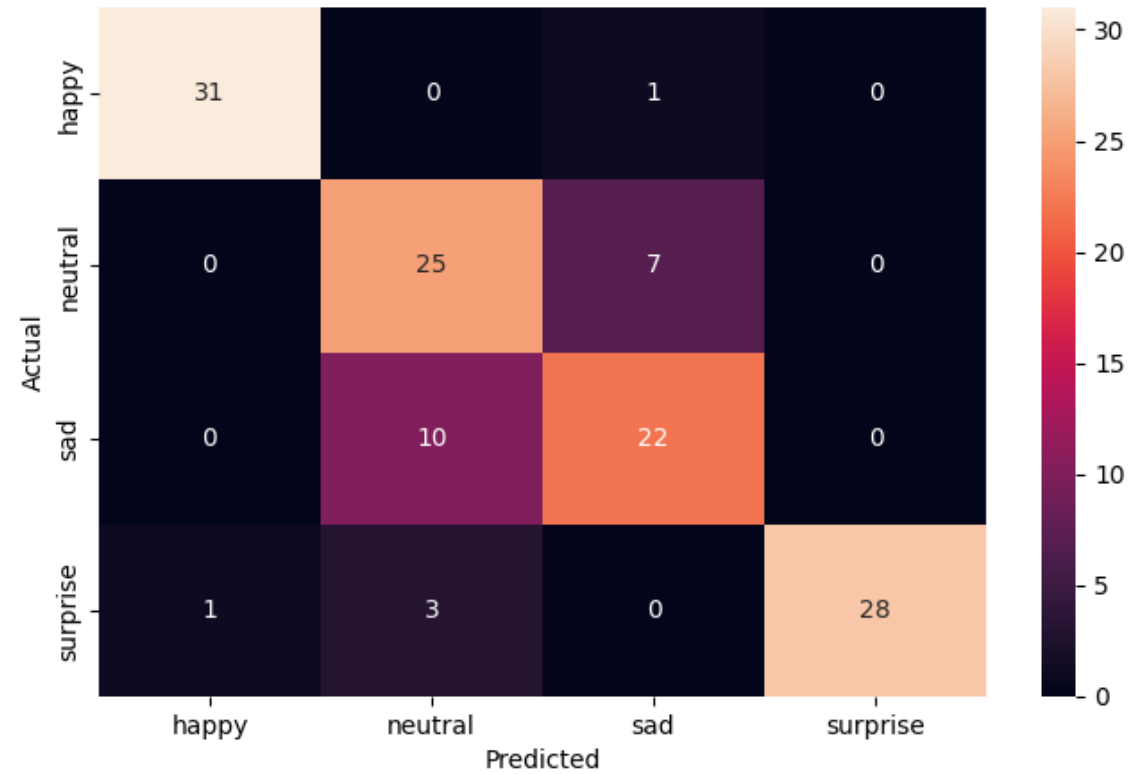


FINAL MODEL SOLUTION

- Model5 had the best performance out of all the models compared
- The appropriate learning rate was able to predict emotions with accuracy of 83%
- F1-Score Range:
 - Sad 71% (Min)
 - Happy 97%(Max)



FINAL MODEL TEST RESULT MATRIX



FINAL MODEL PREDICTION ERRORS

actual: happy
predicted: sad
probability:
0.472



actual: neutral
predicted: sad
probability:
0.583



actual: neutral
predicted: sad
probability:
0.488



actual: neutral
predicted: sad
probability:
0.613



actual: neutral
predicted: sad
probability:
0.79



actual: neutral
predicted: sad
probability:
0.621



actual: neutral
predicted: sad
probability:
0.614



actual: neutral
predicted: sad
probability:
0.913



actual: sad
predicted: neutral
probability:
0.65



actual: sad
predicted: neutral
probability:
0.627



actual: sad
predicted: neutral
probability:
0.943



actual: sad
predicted: neutral
probability:
0.838



actual: sad
predicted: neutral
probability:
0.714



actual: sad
predicted: neutral
probability:
0.719



actual: sad
predicted: neutral
probability:
0.471



actual: sad
predicted: neutral
probability:
0.925



actual: sad
predicted: neutral
probability:
0.89



actual: sad
predicted: neutral
probability:
0.807



actual: surprise
predicted: neutral
probability:
0.496



actual: surprise
predicted: happy
probability:
0.854



actual: surprise
predicted: neutral
probability:
0.591



actual: surprise
predicted: neutral
probability:
0.834



PROPOSED BUSINESS SOLUTION

- Data Enhancement to improve accuracy
- Model Maintenance and Deployment
- Multimodal Emotion Recognition
- Develop Policy to recommend
 - Routine Mental Health Evaluation

EXECUTING BUSINESS SOLUTION

- Image Data Improvement
 - Verification
 - Variety (Age, Race, Gender)
 - Quality (224x224)
 - Color
- Model Maintenance
 - Deployment to Cloud
 - Tuning
 - Dynamic Environment
- Investigate Speech Emotion
- Provide Talking and/or Medication Therapy

EXECUTIVE SUMMARY

- Convolutional Neural Network Model can predict facial emotions with appropriate parameters.
- Facial Emotion Recognition (FER) can help identify potential patients for treatment.
- Proper management of Depression can help reduce the associated economic loss.
- Investigate Multi Modal Emotion Detection

FACIAL EMOTION DETECTION

RISKS

Misinterpretation

- Neutral as Sad (False Positive)
 - increases the cost of unnecessary treatment
- Sad as Neutral (False Negative)
 - prevents needed treatment

FACIAL EMOTION DETECTION

CHALLENGES

- Distinguish Sad and Neutral
- Image Quality and variety
- Availability of treatment plans
- Social Stigma
- Privacy concerns



THANK YOU



APPENDIX

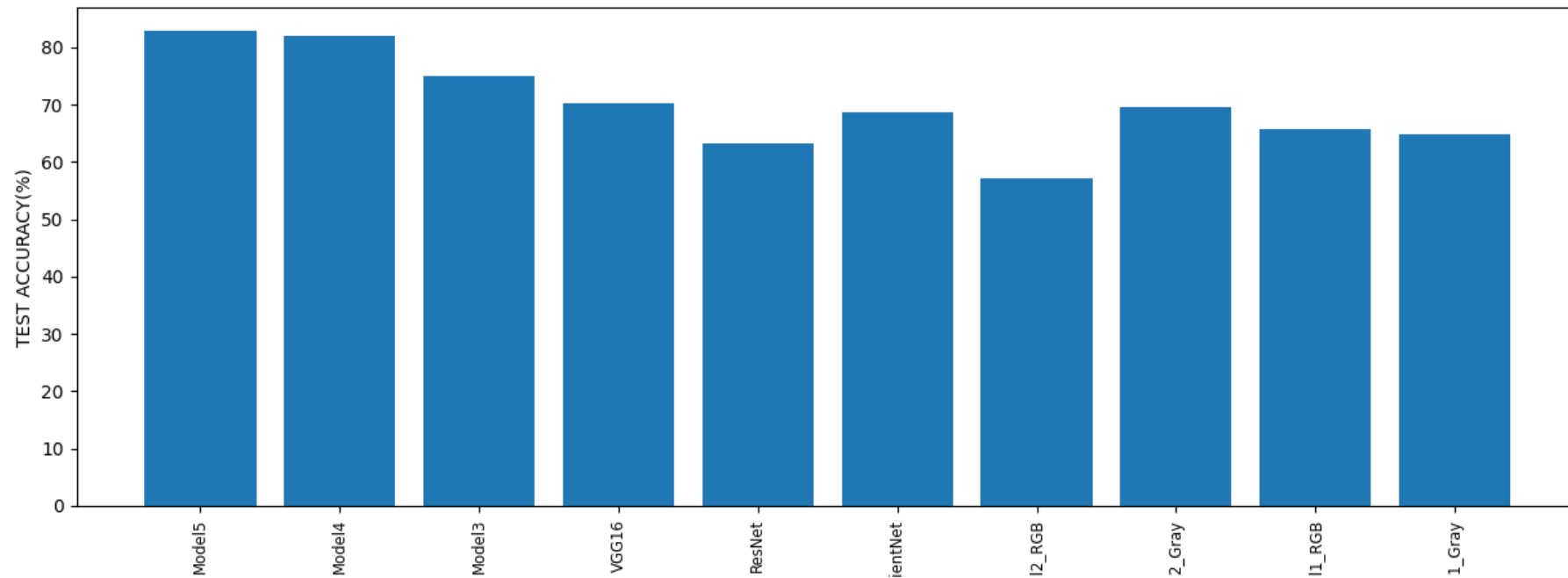
Model Design Iteration (contd.)

Iteration	Model 3	Model 4	Model 5
Training Accuracy	Underfitting	Overfitting	Improved
5 th Conv. Layer	128	256	512
Conv. Filter (Kernel)	2x2	2x2	3x3
ADAM LR	0.003	0.002	0.002

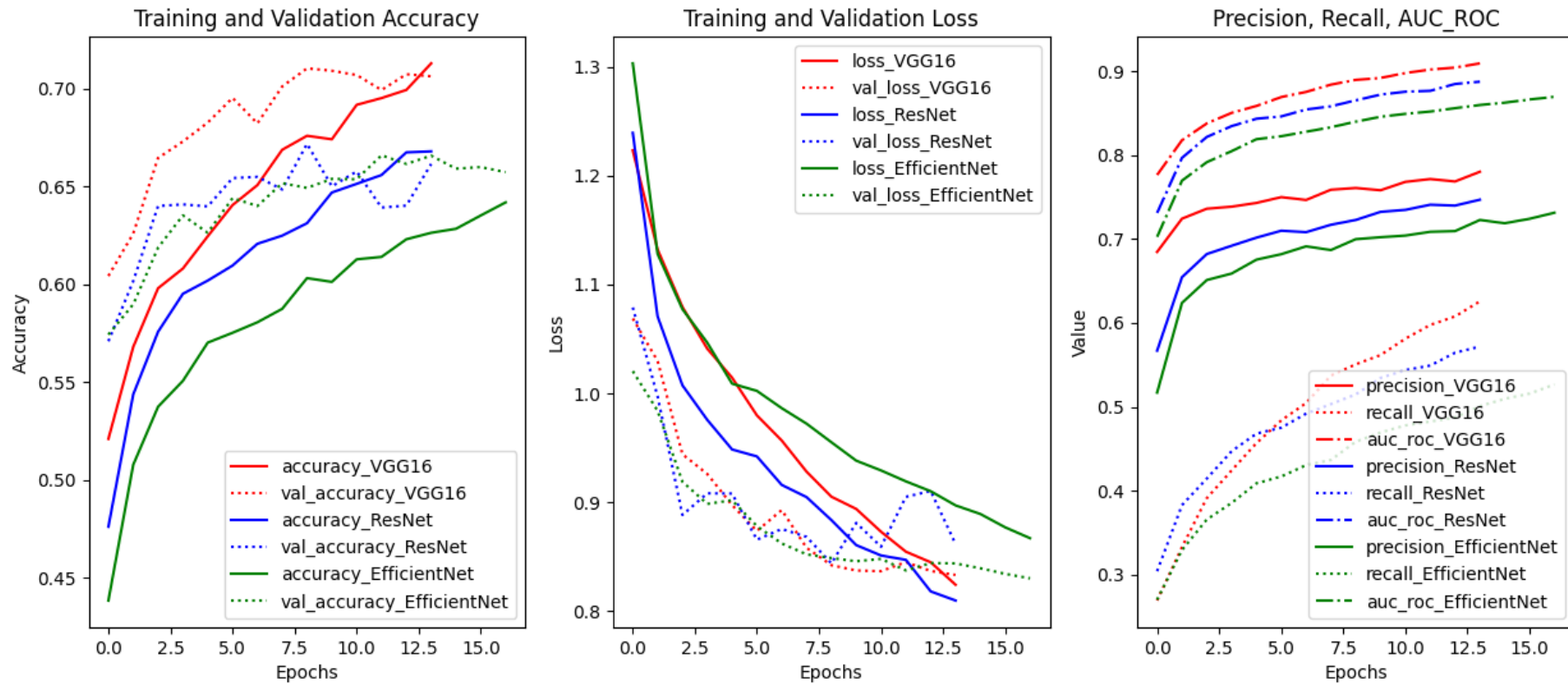
Model Performance Metric Comparison

	vgg16	resNet	EfficientNet	model3	model4	model5
loss	82.63	81.87	80.22	55.216	47.792	50.729
accuracy	70.31	63.28	68.75	75.000	82.031	82.812
precision	78.10	73.47	74.74	80.357	82.787	85.000
recall	64.06	56.25	55.47	70.312	78.906	79.688
auc_roc	90.99	88.24	88.85	94.717	96.142	95.923

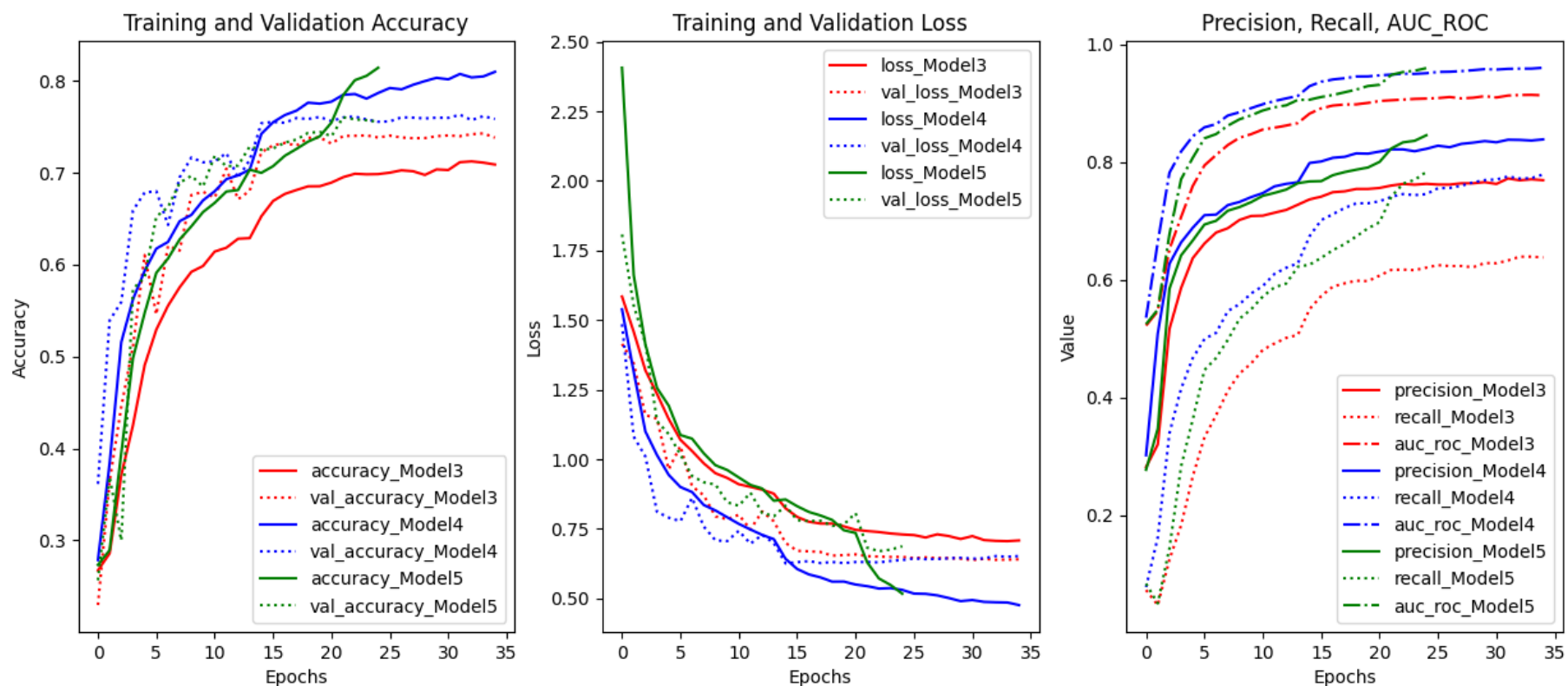
MODEL COMPARISON



Transfer Model Comparison



Complex CNN Model Design Iteration



Gray scale vs RGB Comparison

